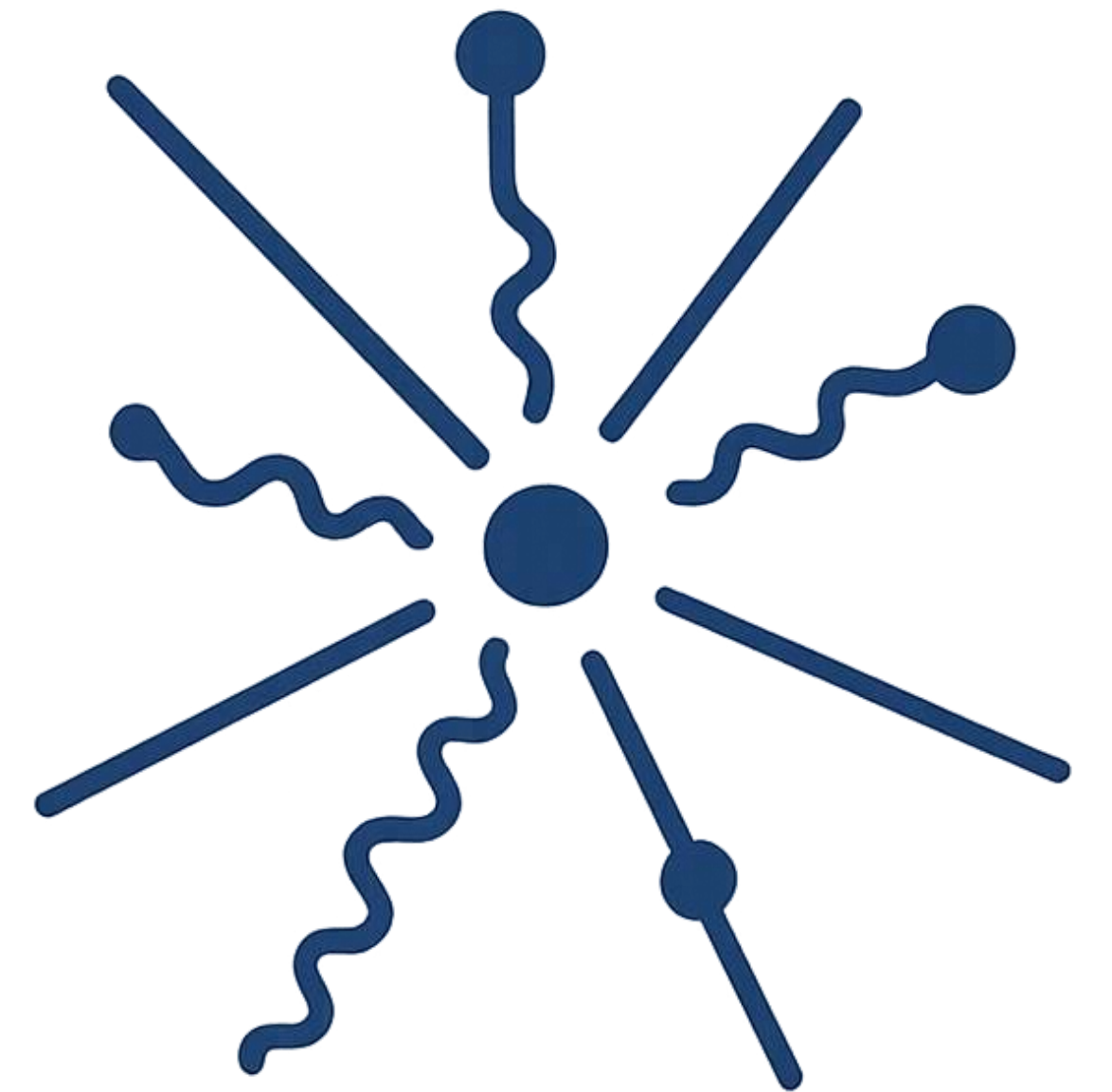
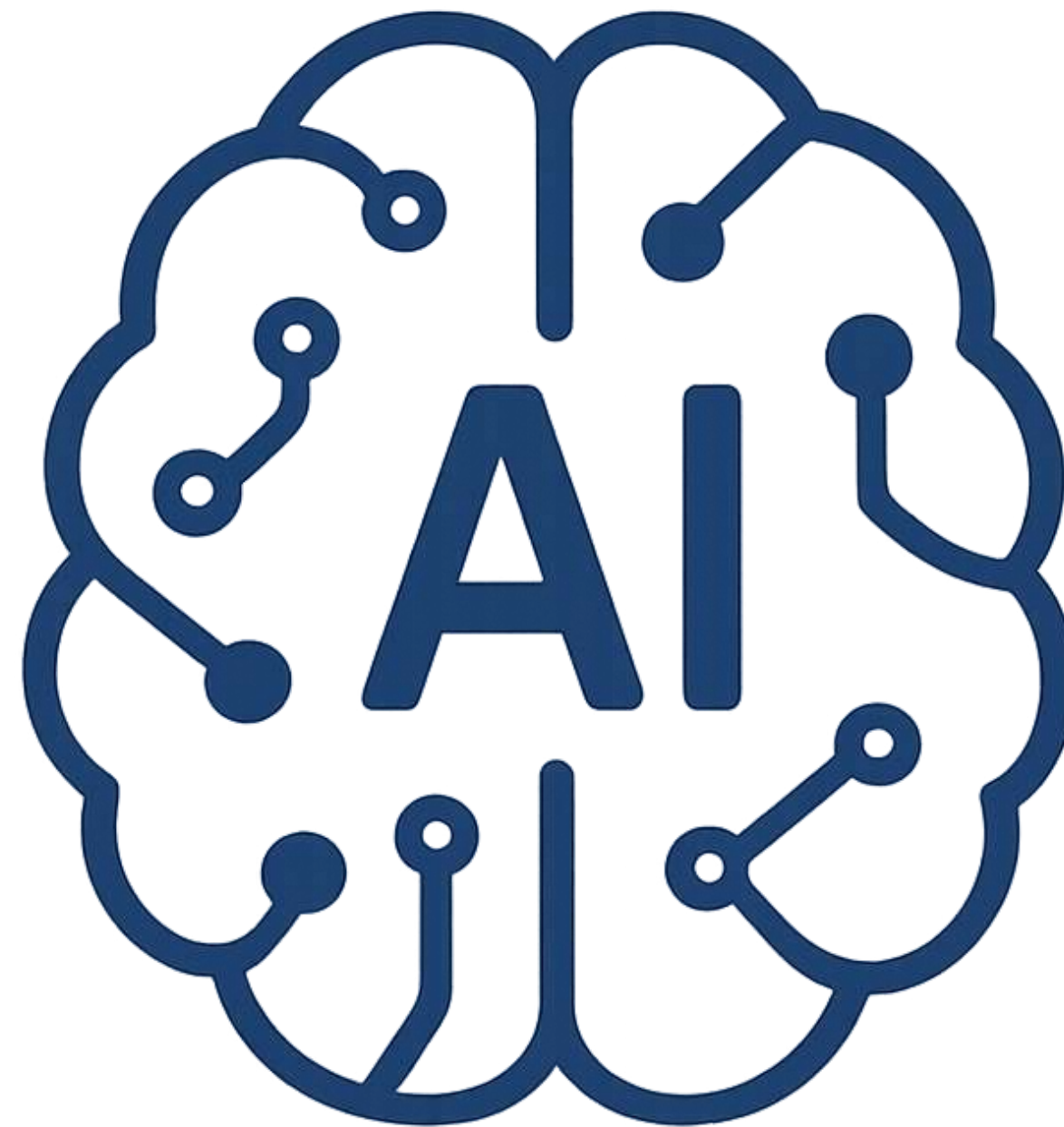


Introduction to

AI-Driven HEP

S. A. Fard - School of Physics (IPM)





AI-Driven HEP 15



Session 15

Representational Learning



Reference

[1] Bishop, Christopher M., and Hugh Bishop. Deep learning: Foundations and concepts. Springer Nature, 2023

<https://physics.ipm.ac.ir/~ansarifard/>





Realm of Meaning

Describe the image semantically





Realm of Meaning

Describe the image semantically

... a Renaissance-style portrait of a woman reminiscent of the Mona Lisa, posed with folded hands and a calm, subtly enigmatic expression. Her face, hair, and clothing are rendered in warm, soft tones using sfumato-like shading. The background is ...

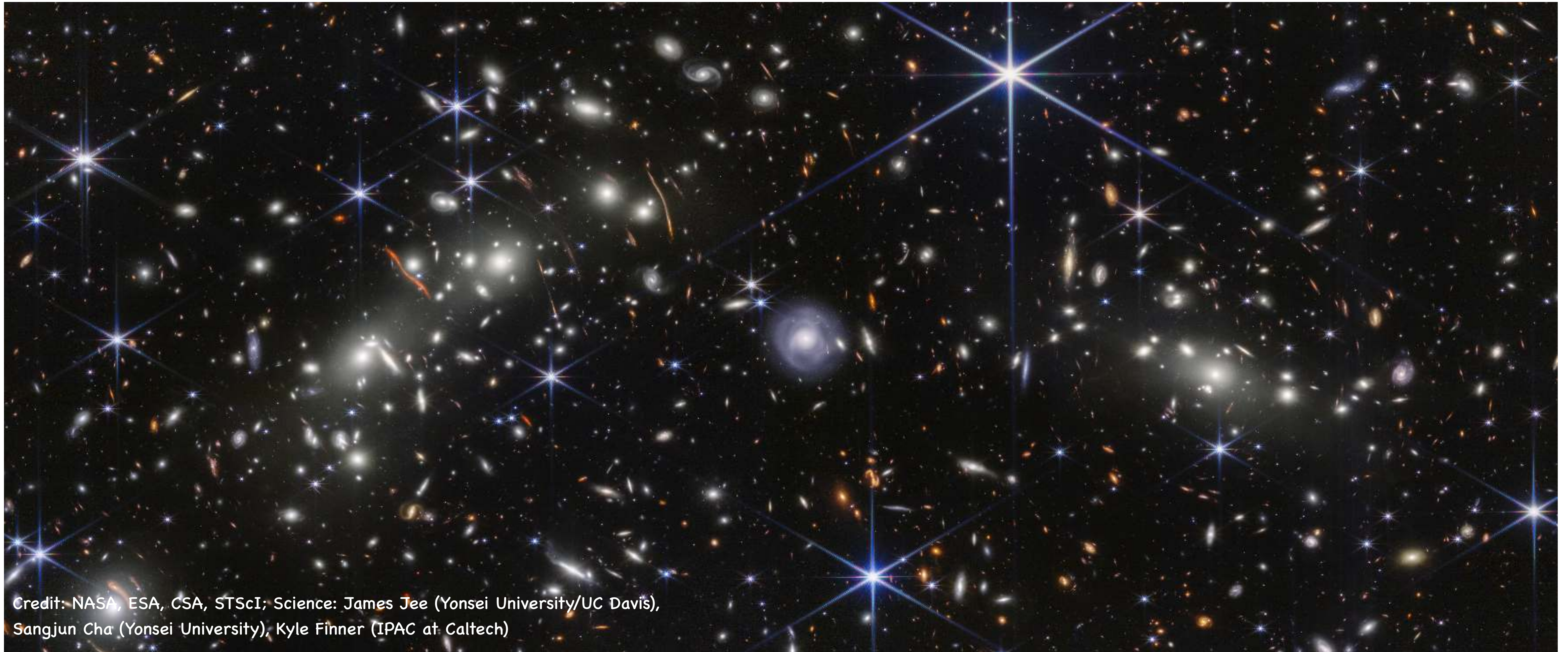




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Realm of Meaning

Describe the image semantically ?!



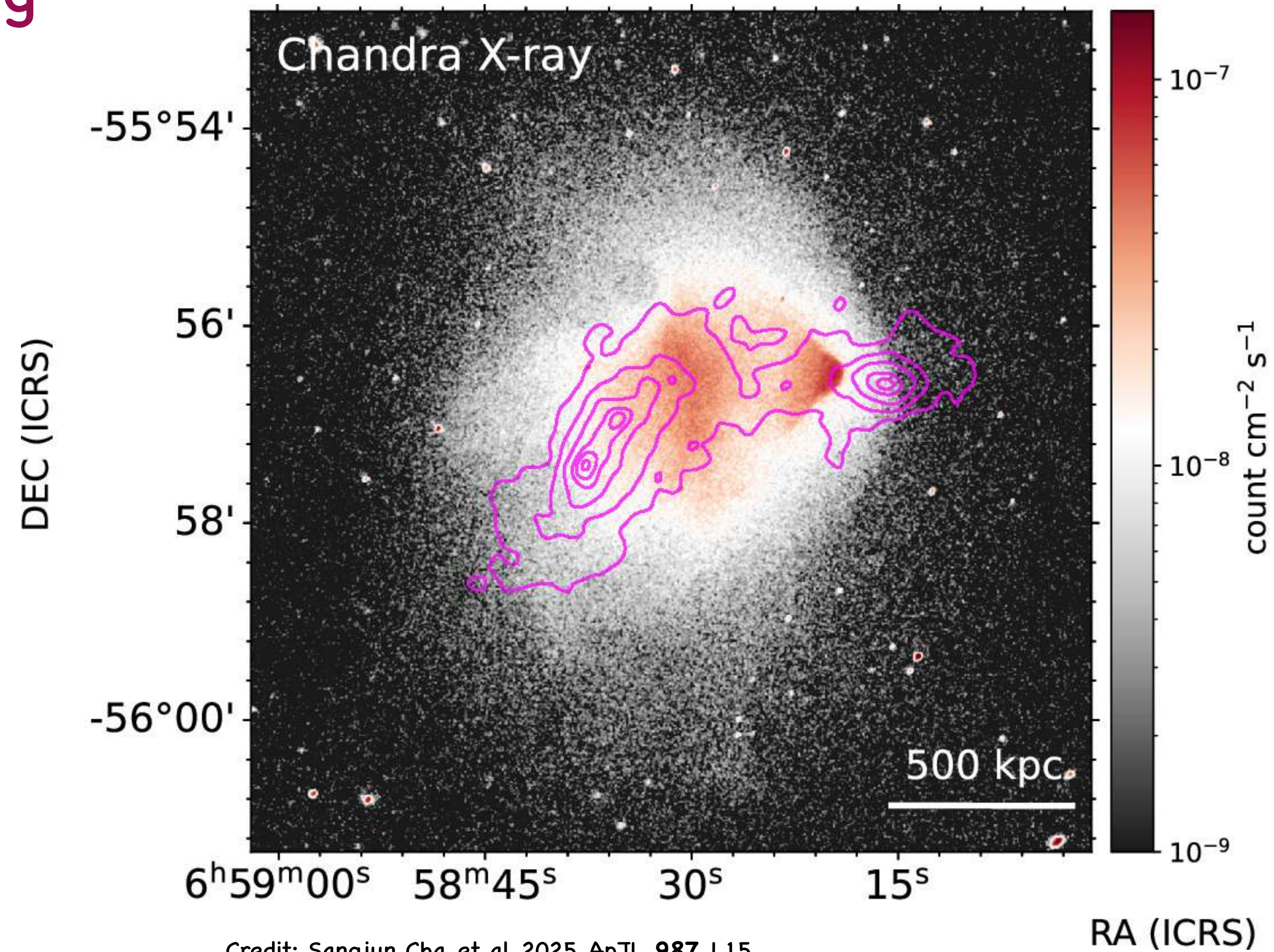
Credit: NASA, ESA, CSA, STScI; Science: James Jee (Yonsei University/UC Davis), Sangjun Cha (Yonsei University), Kyle Finner (IPAC at Caltech)

Realm of Meaning





Realm of Meaning

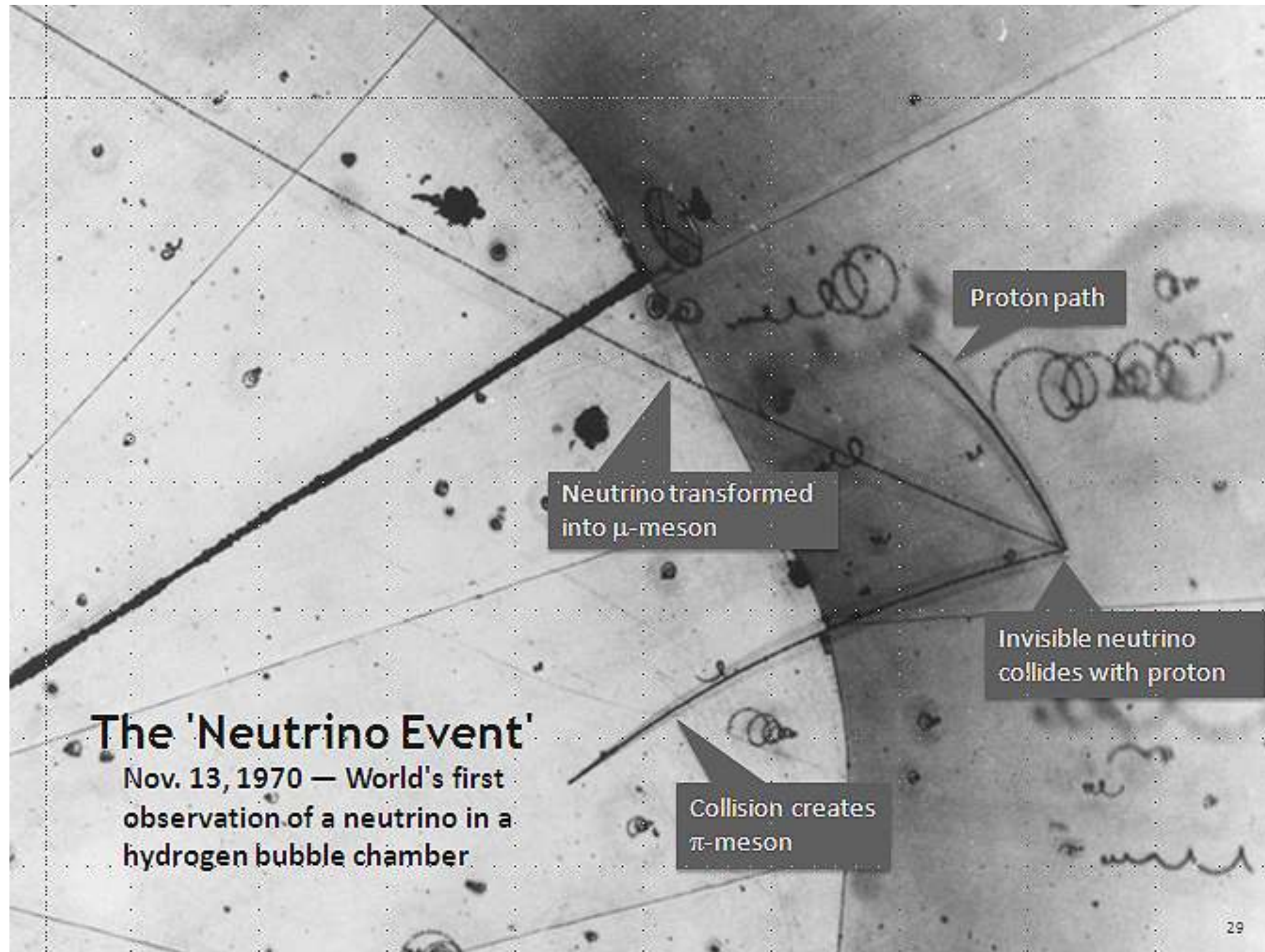




AI-Driven HEP 15

Realm of Meaning

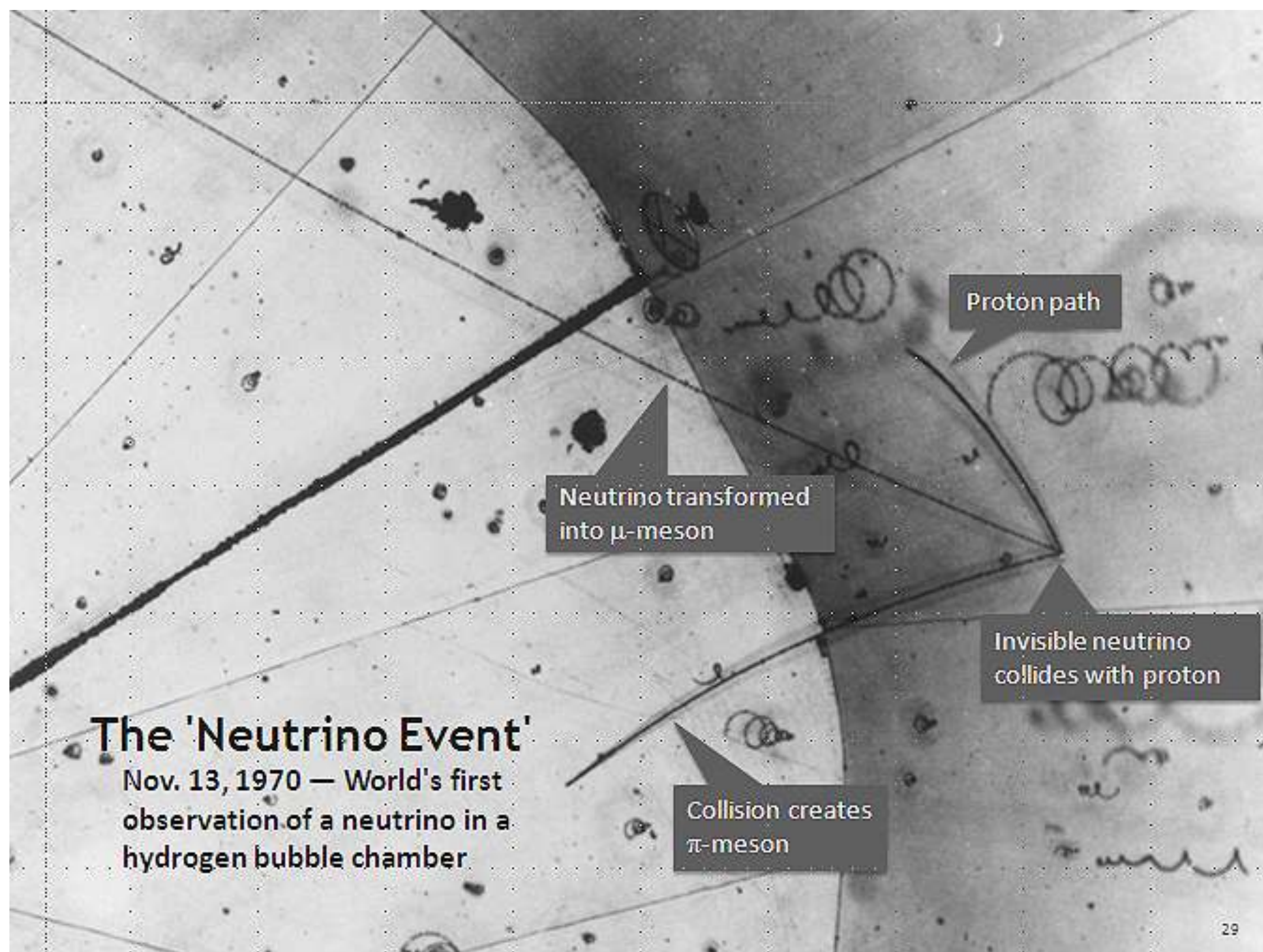
Describe the image semantically ?!





Realm of Meaning

Describe the image semantically ?!



When a **muon neutrino** (ν_μ) hits a **proton** (p), it can interact via the **charged-current weak interaction** by exchanging a W^+ boson.

The W^+ turns the neutrino into a **muon** (μ^-) and converts the proton's **up quark** into a **down quark**, creating a **neutron**.

The struck quark is left with enough energy that the system produces an additional hadron — typically a **pion**.

Physical picture:

- The neutrino transfers energy and momentum to a quark inside the proton.
- The weak W^+ exchange changes flavor.
- The final-state quarks hadronize, producing a **pion** (π^+) along with the **muon**.

This process is called **charged-current single pion production (CC1 π)** and is important in many neutrino experiments.



Hierarchical Representations



sequence of internal transformations

$$x \rightarrow h^{(1)} \rightarrow h^{(2)} \rightarrow \dots \rightarrow h^{(L)} \rightarrow y$$

Each layer extracts a **representation** of the input

- **Early layers** learn local, simple features
(edges, textures in CNNs; short patterns in NLP).
- **Middle layers** learn combinations of features
(corners, shapes, semantic parts).
- **Deep layers** learn high-level concepts
(objects, categories, sentiment).



Distributed Representations



Each concept is represented by many neurons

a pattern of activations across many units encodes concepts

If each of N neurons can be in 2 states (on/off), you can represent 2^N distinct concepts.

represent an exponential number of patterns



Representation Learning

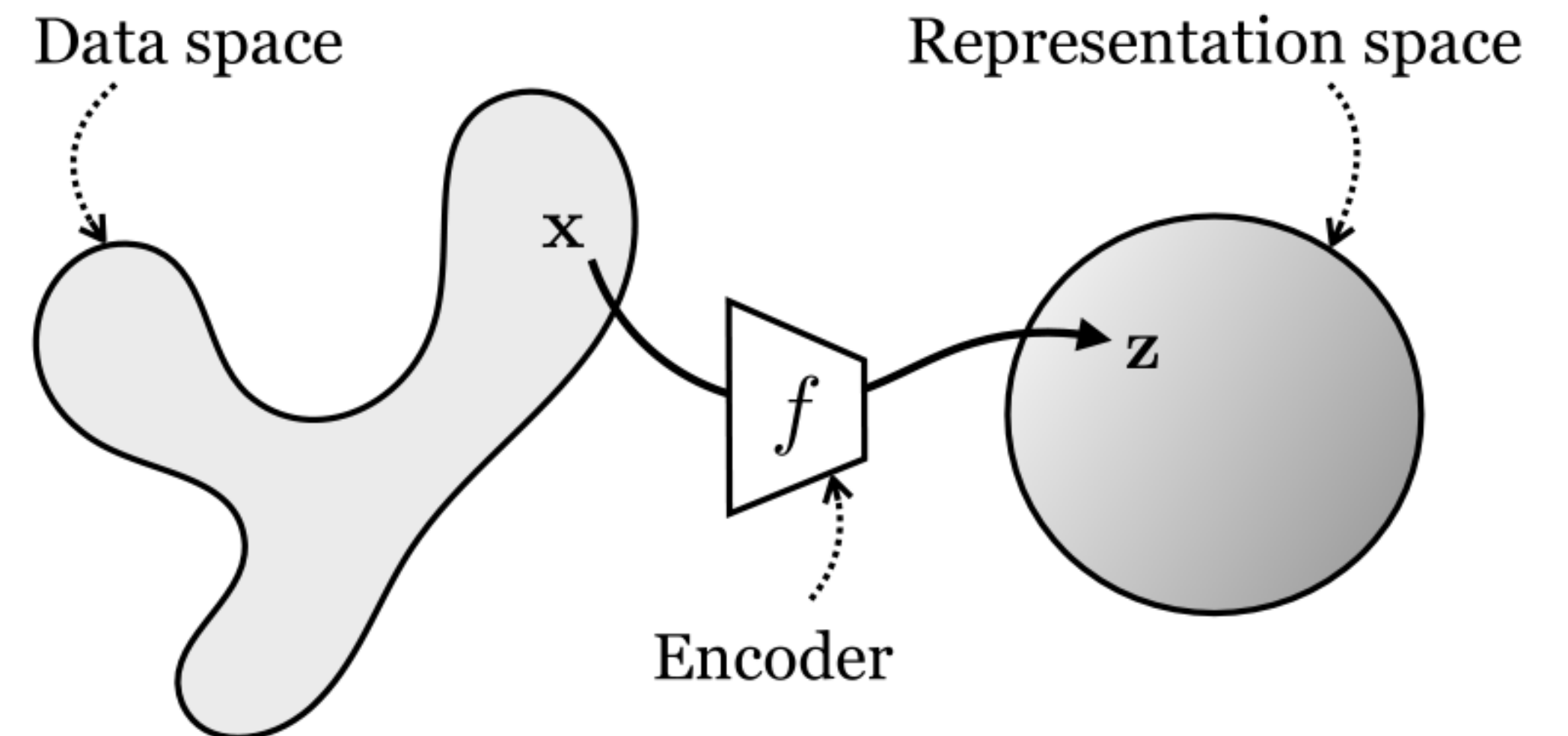


learning **nonlinear transformations** of the input that converts raw data (pixels, words, signals) into a new form

makes tasks **much easier**

A representation sometimes called embedding

representation puts similar inputs close together in the embedding space



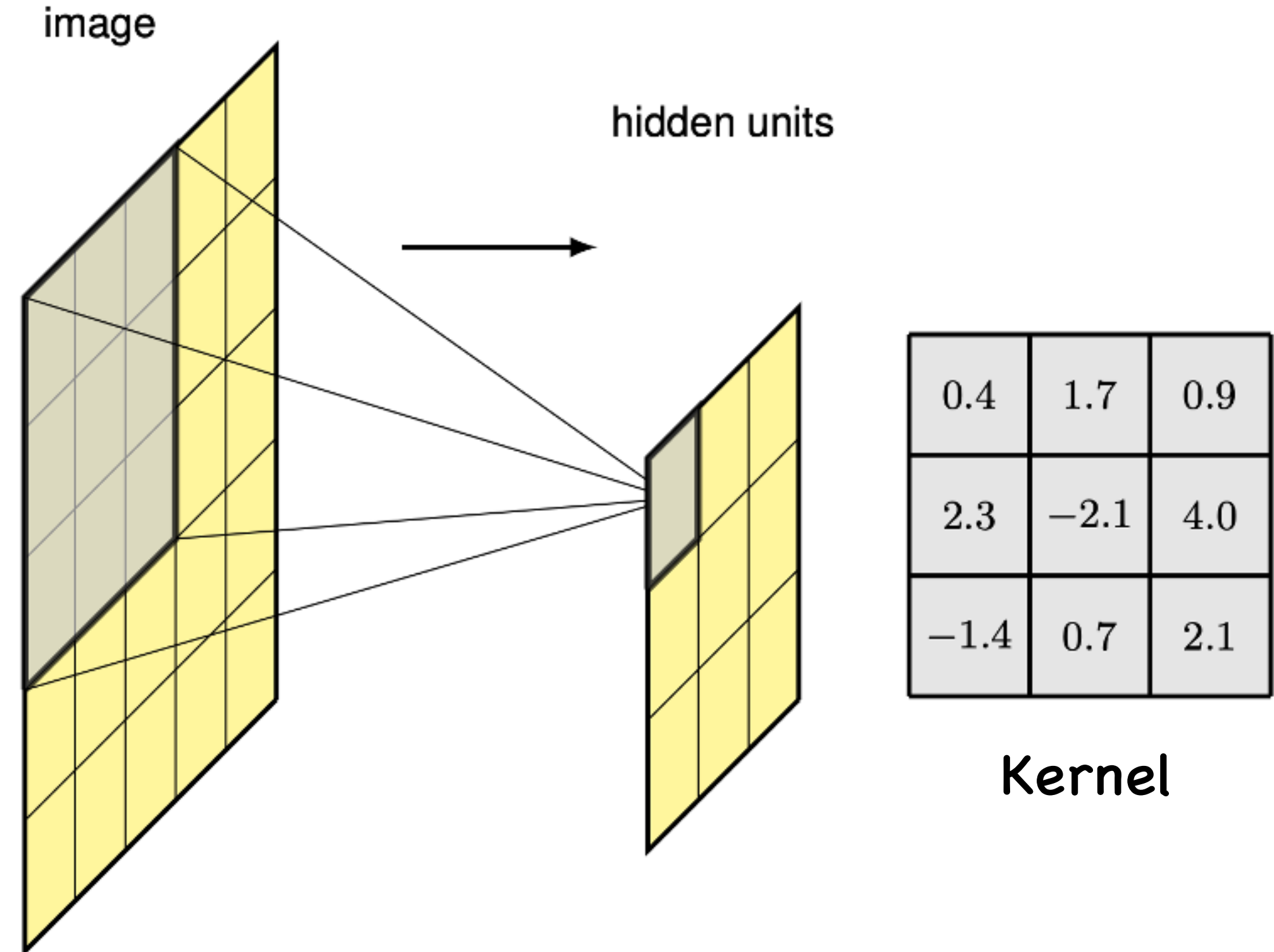


Representational Learning

Network's ability to automatically discover and learn the optimal set of features from raw image

The patch of image is refer as the receptive field of hidden unit

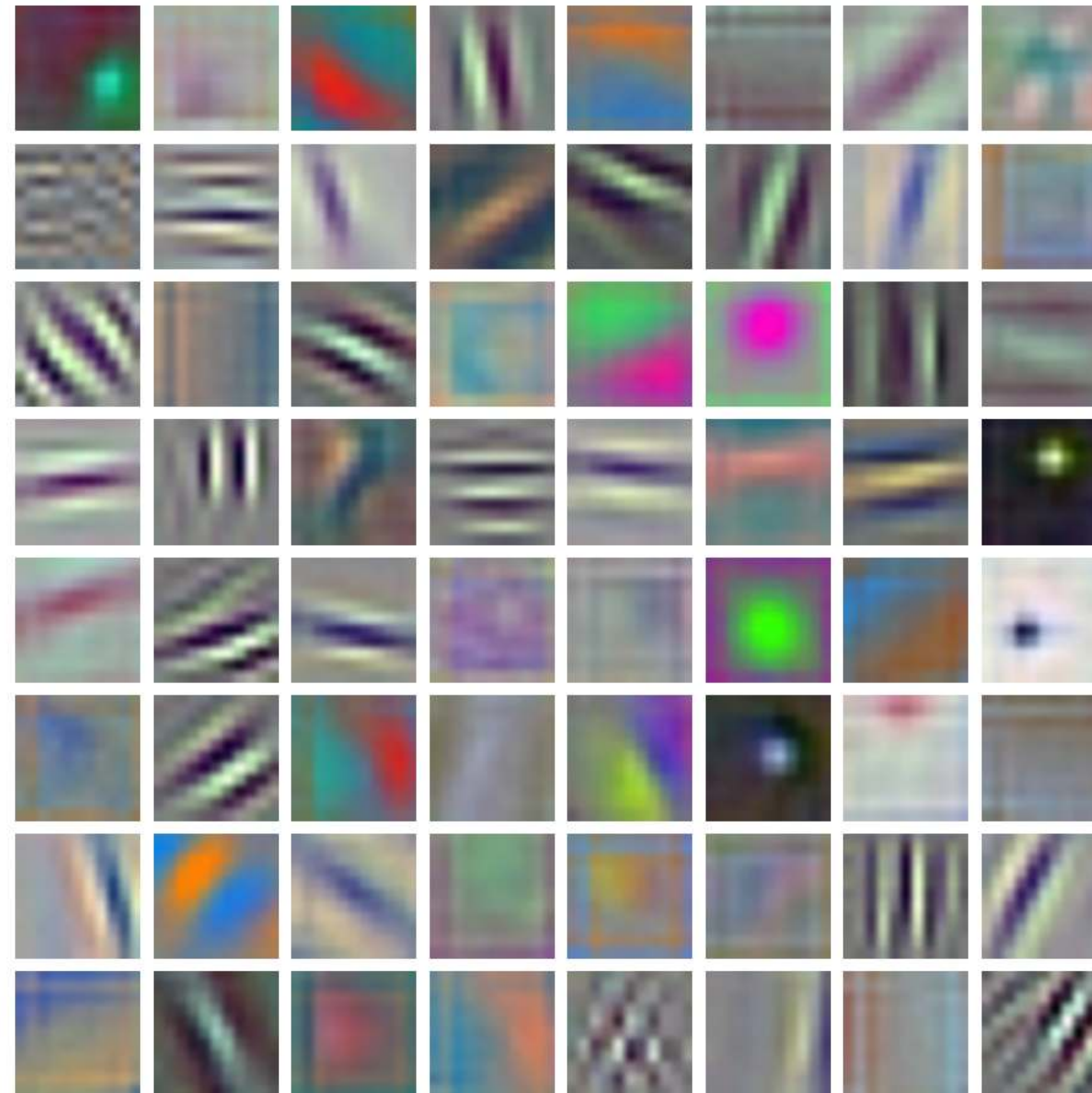
$$z = \text{ReLU}(\mathbf{w}^T \mathbf{x} + w_0)$$





Representational Learning

Visualize the kernels in the first layer of a CNN

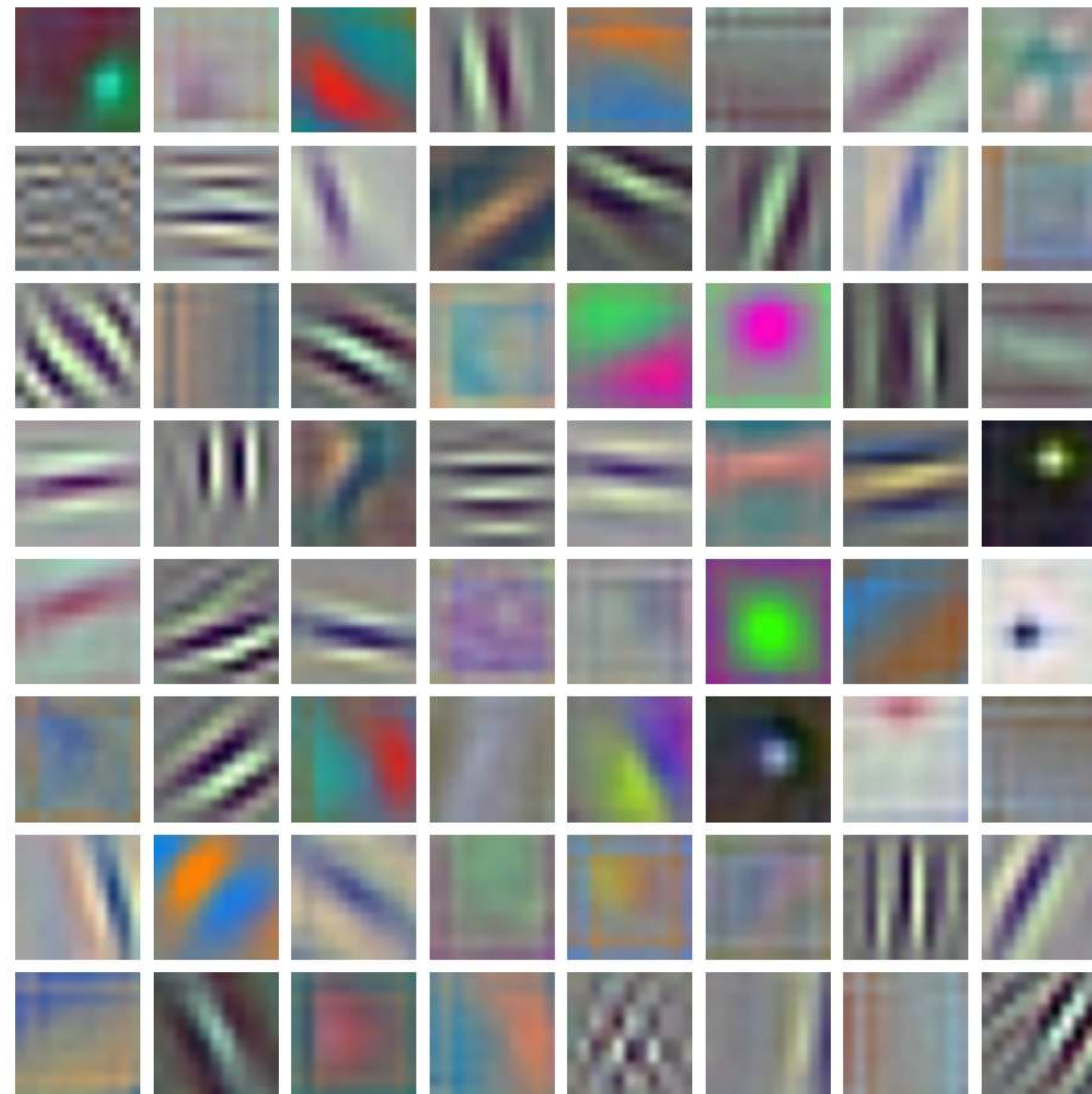


Extract the low level features of images



Representational Learning

Visualize the kernels in the first layer of a CNN



Extract the low level features of images

The subsequent layers in the network are harder to interpret



Semantic Segmentation

The importance of meaning or logic



In daily life it is related to natural objects



Semantic Segmentation

The importance of meaning or logic



In daily life it is related to natural objects

What is the correspondence in physics?



Semantic Segmentation

The importance of meaning or logic



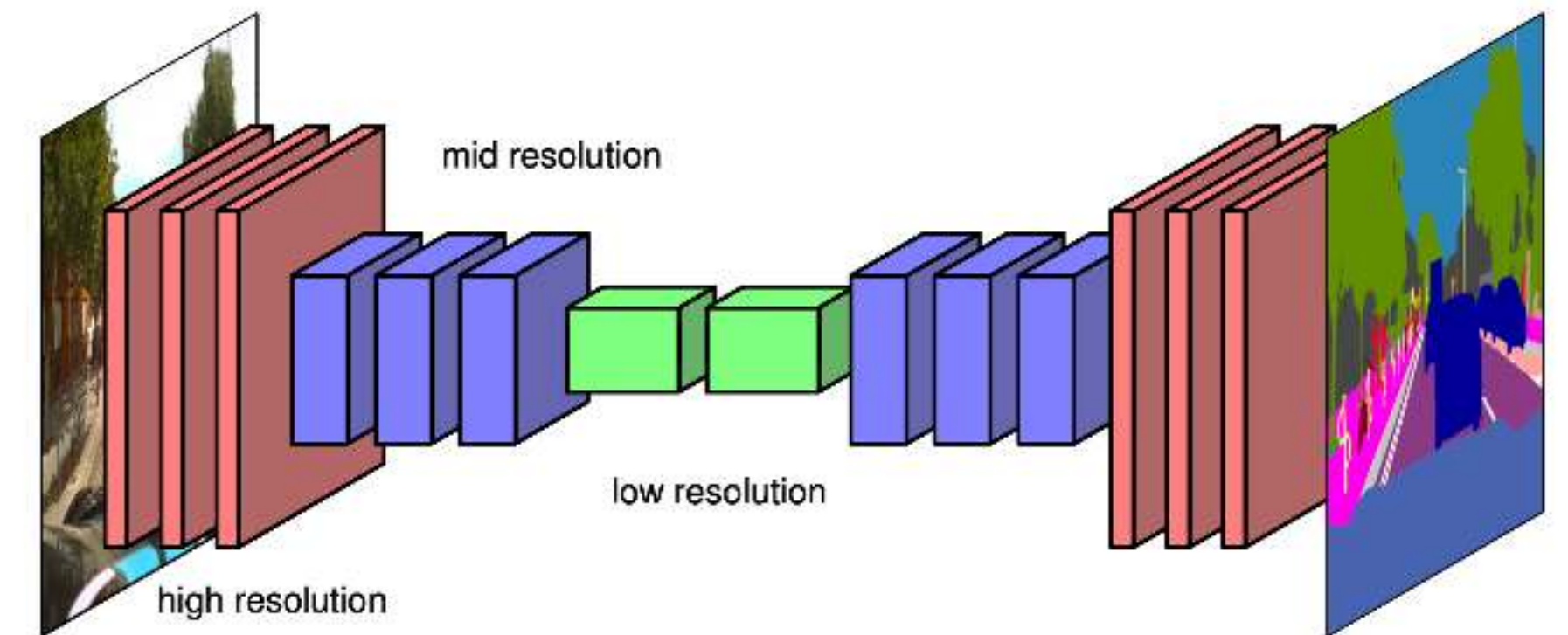
In daily life it is related to natural objects

What is the correspondence in physics ? Identify and label regions with distinct physical properties
Values of physical quantities (e.g., temperature, density, potential)



Semantic Segmentation

The importance of meaning or logic



In daily life it is related to natural objects

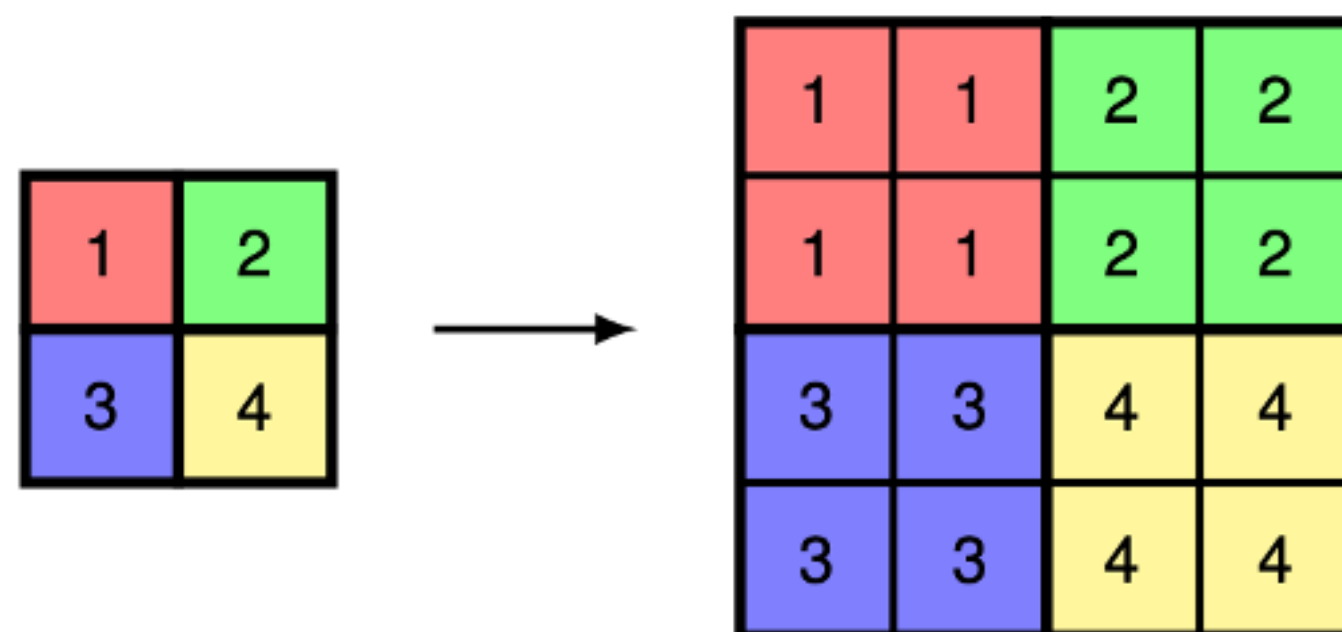
What is the correspondence in physics ? Identify and label regions with distinct physical properties
Values of physical quantities (e.g., temperature, density, potential)



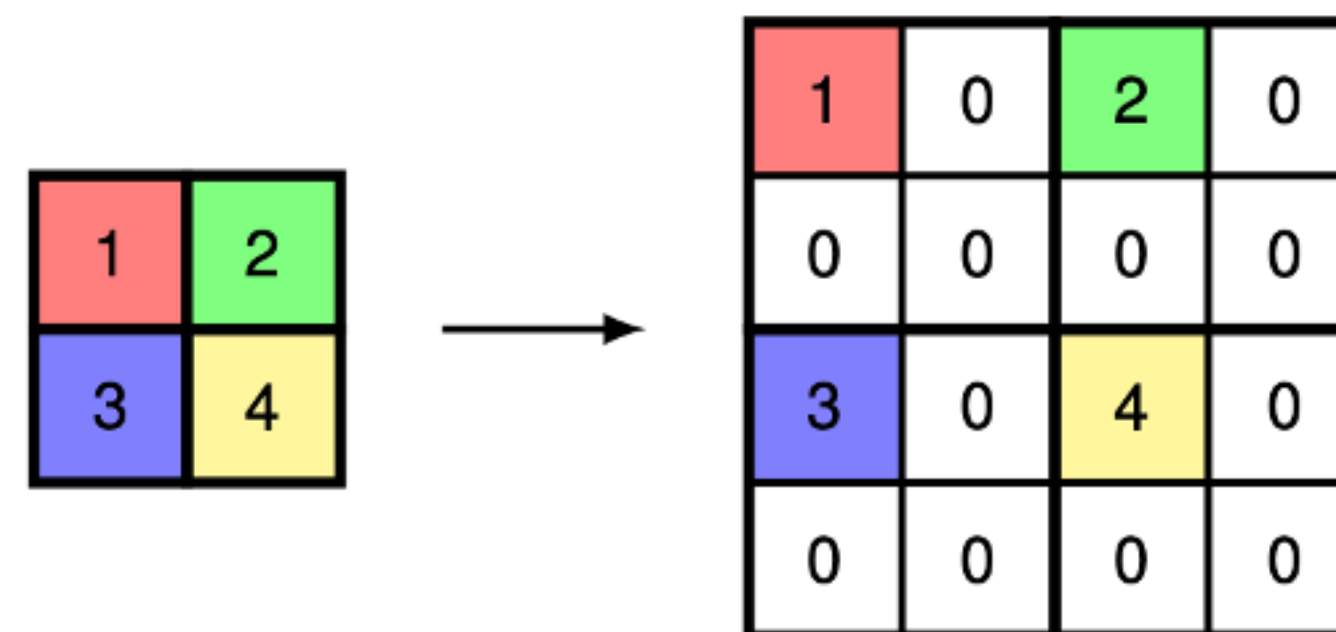
Up-Sampling

The loss of spatial information is a problem for semantic segmentation as we want to classify each pixel

Average or max pooling



(a)



(b)



Transpose Convolution

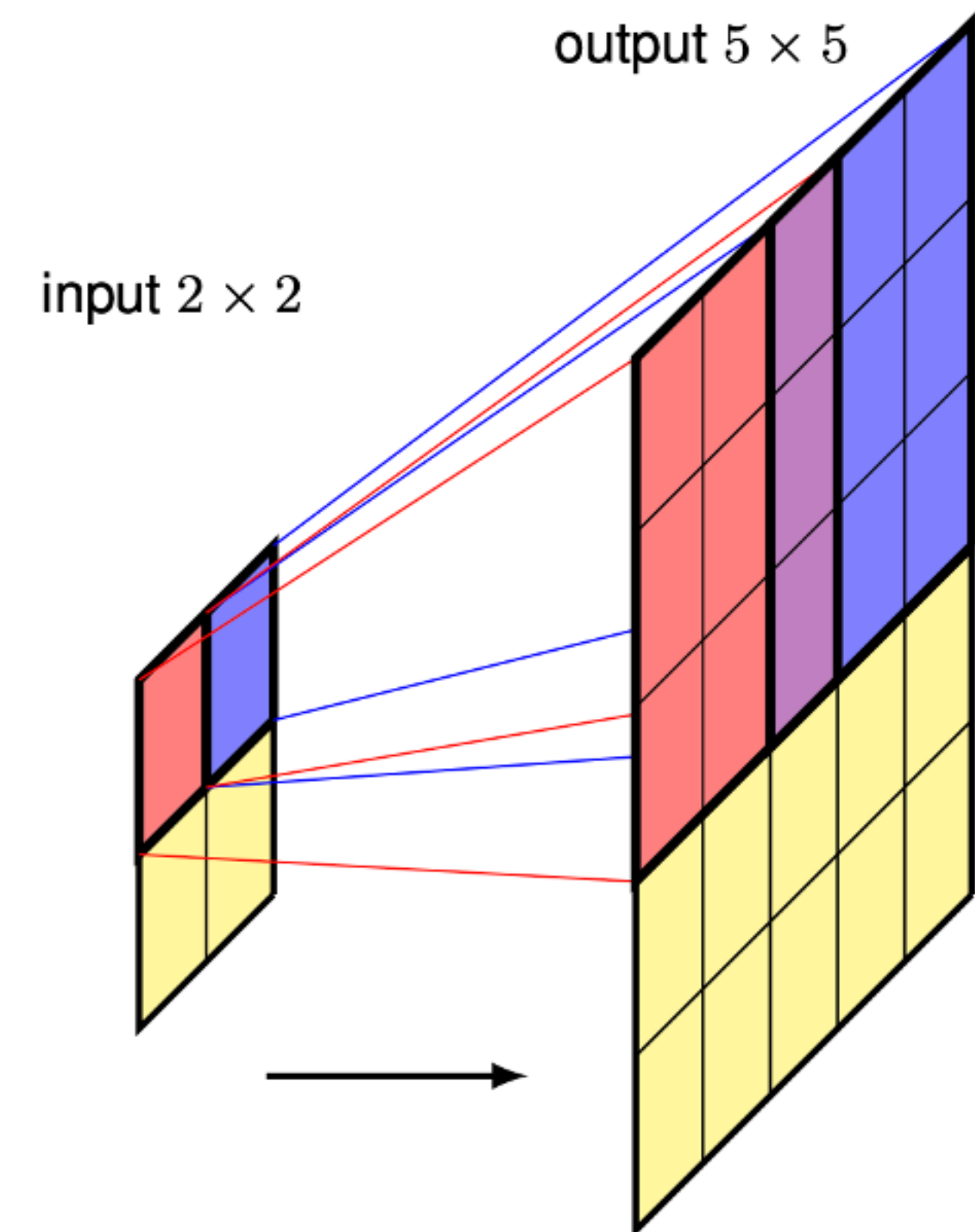
The 3 by 3 kernel with an output stride of 2

[Red, Blue]
[Green, Yellow]

```
[[1, 0, 1],  
 [0, 1, 0],  
 [1, 0, 1]]
```

Output from Red:

```
[1xRed, 0xRed, 1xRed, 0, 0]  
[0xRed, 1xRed, 0xRed, 0, 0]  
[1xRed, 0xRed, 1xRed, 0, 0]  
[0, 0, 0, 0, 0]  
[0, 0, 0, 0, 0]
```





Transpose Convolution

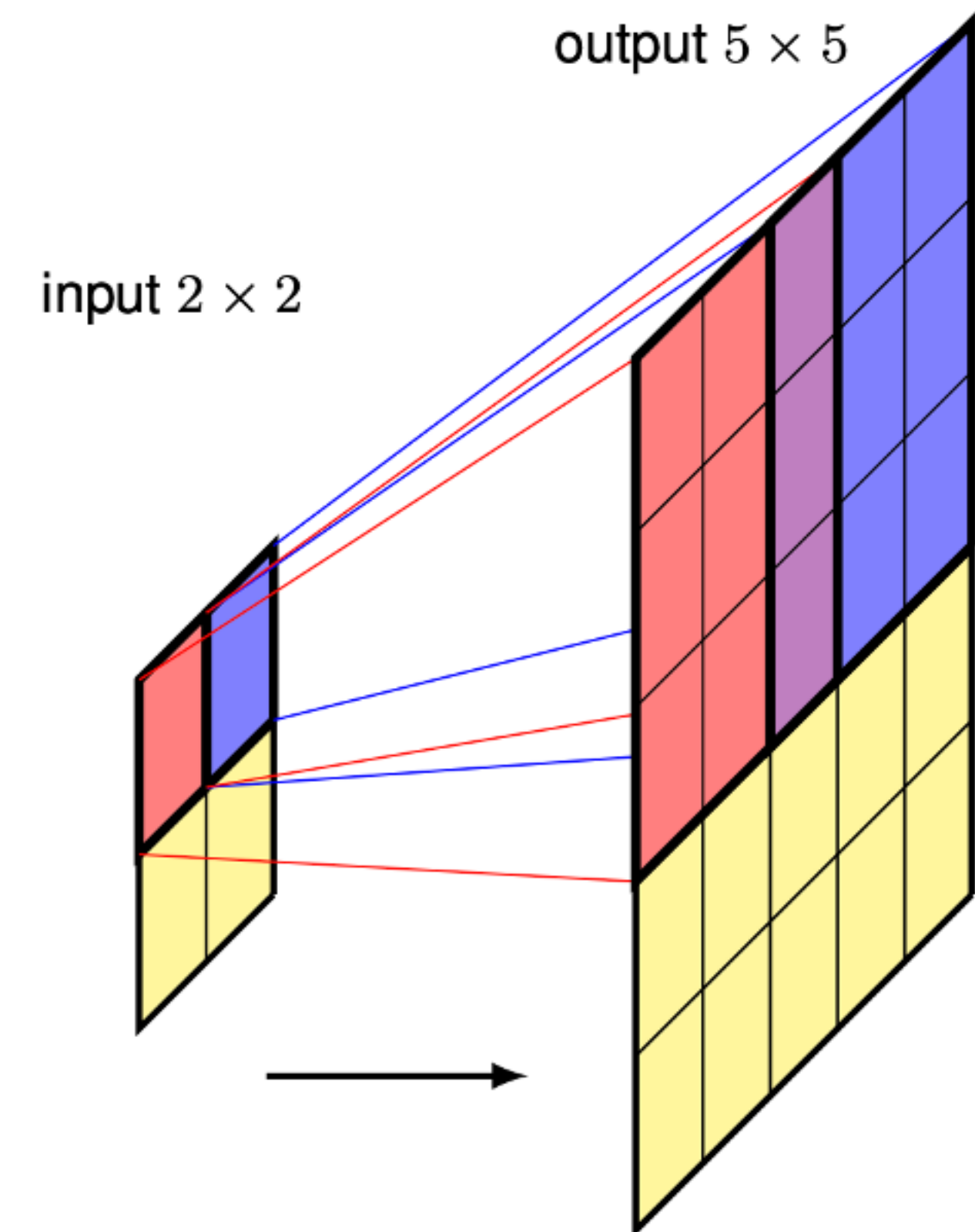
The 3 by 3 kernel with an output stride of 2

```
[Red,  Blue]  
[Green, Yellow]
```

```
[[1, 0, 1],  
 [0, 1, 0],  
 [1, 0, 1]]
```

Output from Blue:

```
[0, 0, 1×Blue, 0×Blue, 1×Blue]  
[0, 0, 0×Blue, 1×Blue, 0×Blue]  
[0, 0, 1×Blue, 0×Blue, 1×Blue]  
[0, 0, 0,      0,      0]  
[0, 0, 0,      0,      0]
```





Transpose Convolution

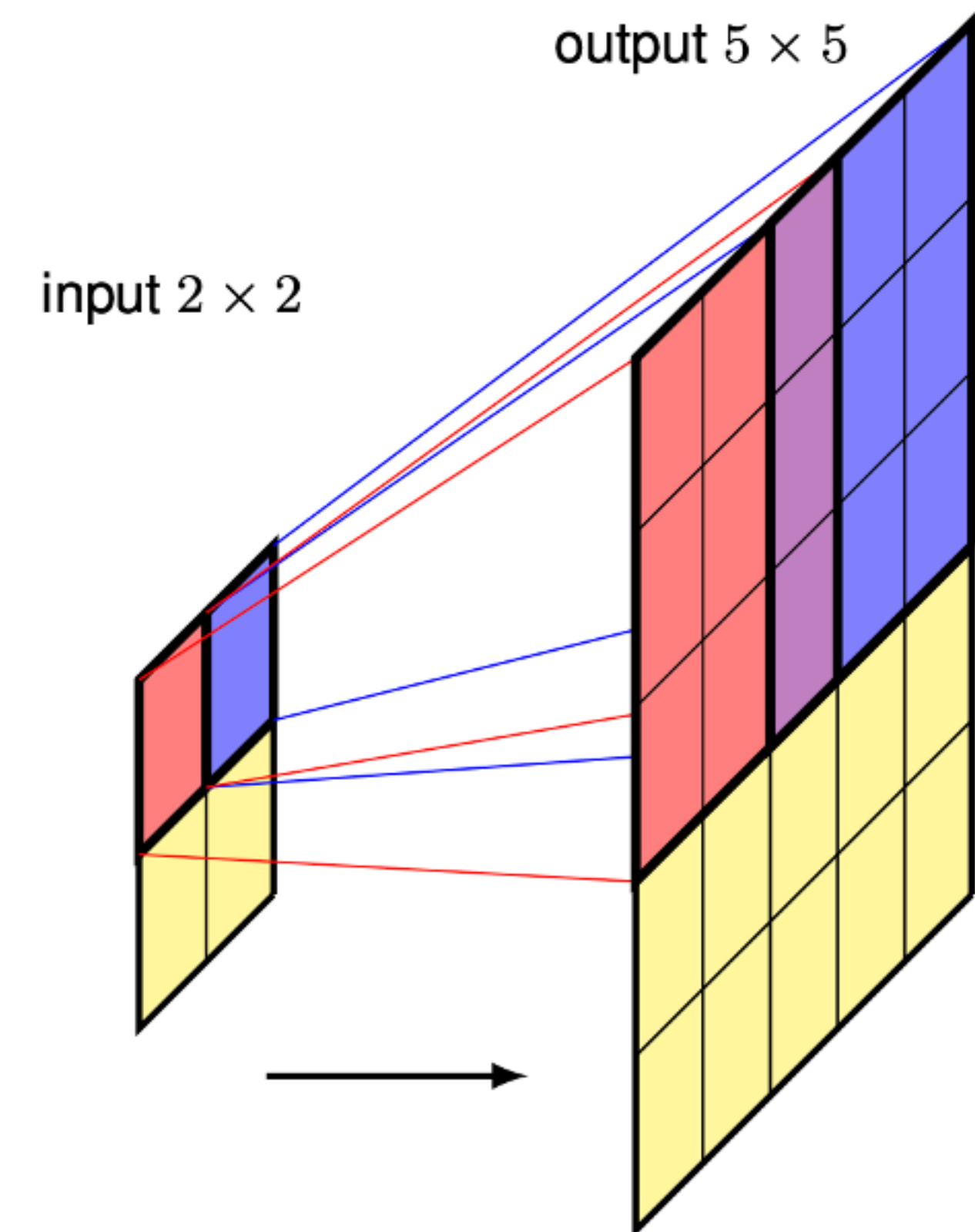
The 3 by 3 kernel with an output stride of 2

```
[Red,   Blue]  
[Green, Yellow]
```

```
[[1, 0, 1],  
 [0, 1, 0],  
 [1, 0, 1]]
```

Output from Green:

```
[0,    0,    0,    0, 0]  
[0,    0,    0,    0, 0]  
[1×Green, 0×Green, 1×Green, 0, 0]  
[0×Green, 1×Green, 0×Green, 0, 0]  
[1×Green, 0×Green, 1×Green, 0, 0]
```





Transpose Convolution

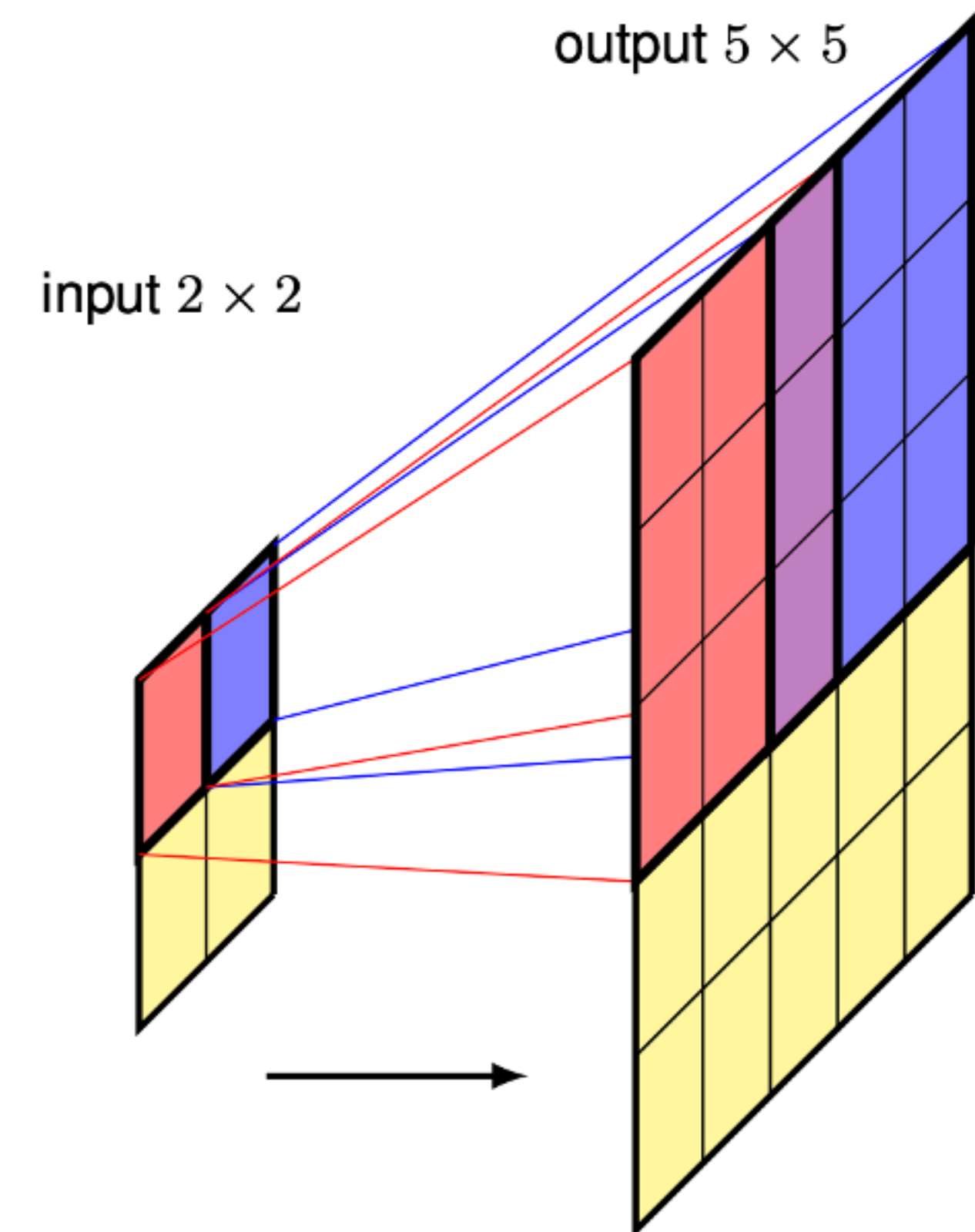
The 3 by 3 kernel with an output stride of 2

```
[Red,   Blue]  
[Green, Yellow]
```

```
[[1, 0, 1],  
 [0, 1, 0],  
 [1, 0, 1]]
```

Output from Yellow:

```
[0, 0, 0,    0,    0]  
[0, 0, 0,    0,    0]  
[0, 0, 1×Yellow, 0×Yellow, 1×Yellow]  
[0, 0, 0×Yellow, 1×Yellow, 0×Yellow]  
[0, 0, 1×Yellow, 0×Yellow, 1×Yellow]
```





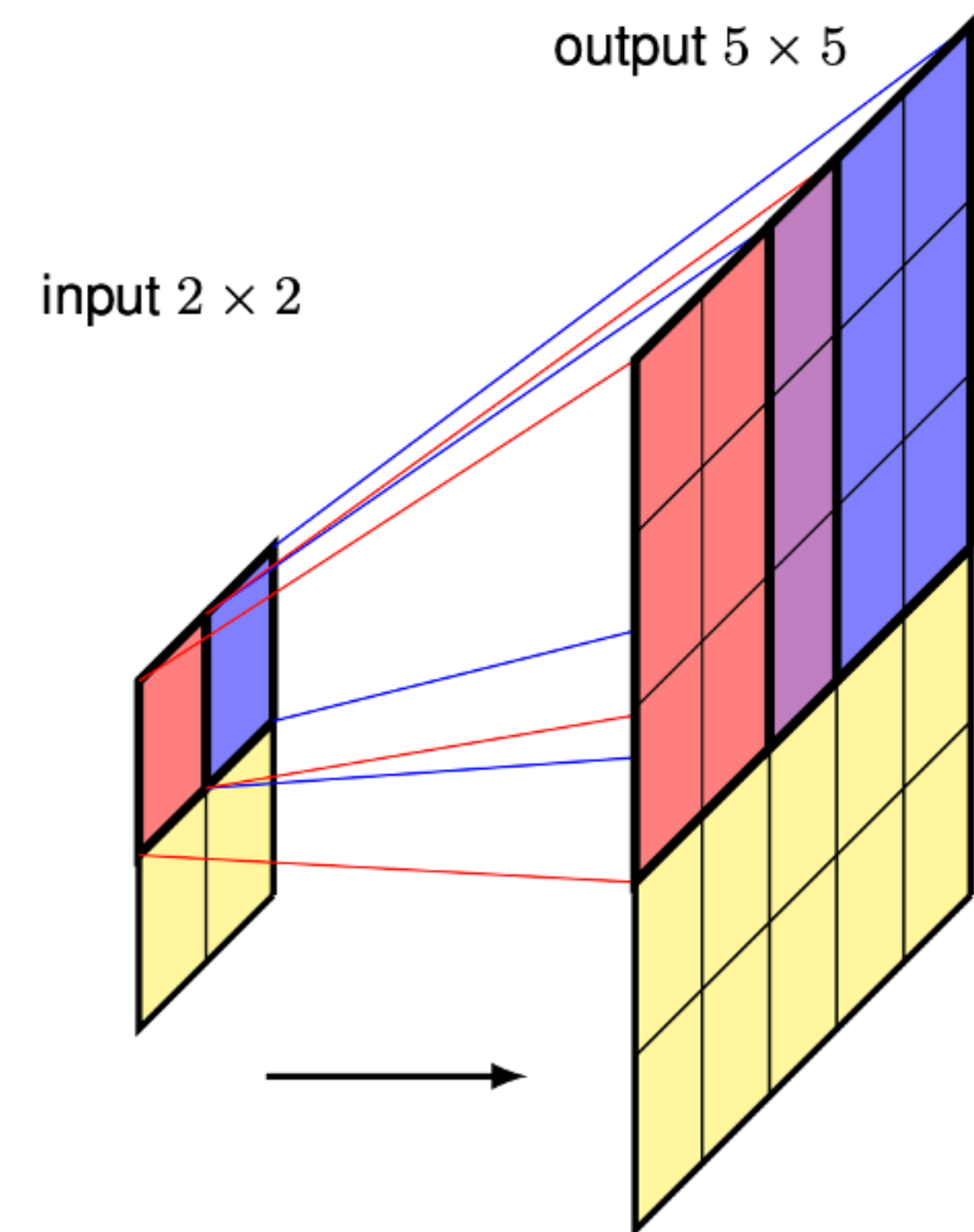
Transpose Convolution

The 3 by 3 kernel with an output stride of 2

```
[Red,  Blue]
[Green, Yellow]
```

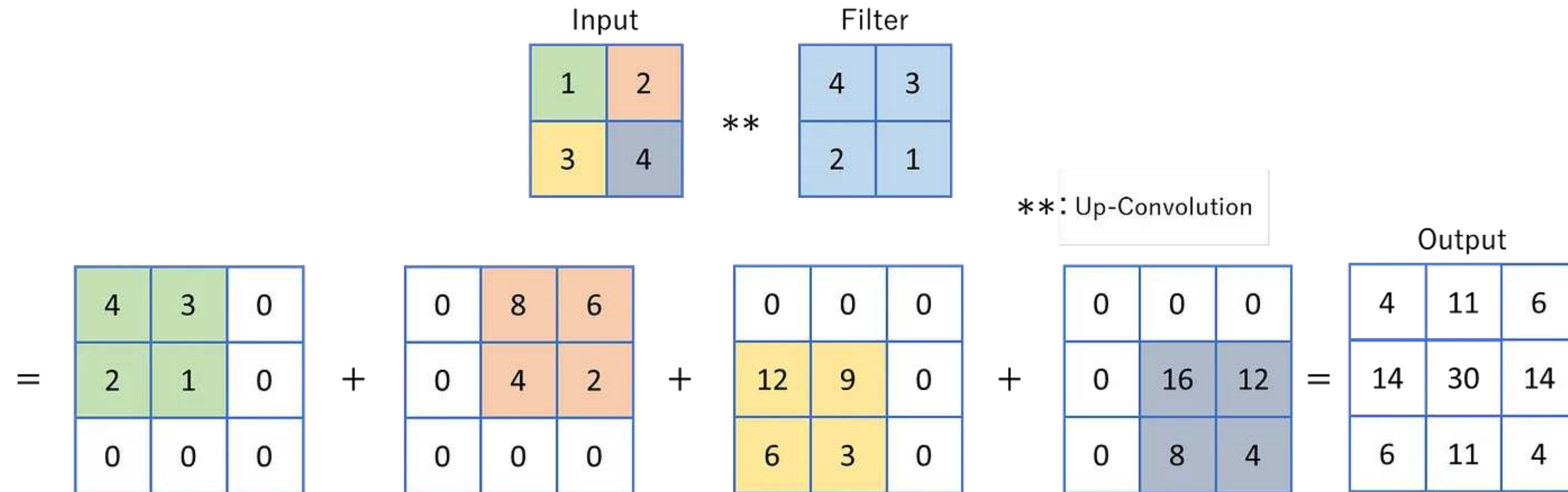
```
[[1, 0, 1],
 [0, 1, 0],
 [1, 0, 1]]
```

```
[Red,  0,  Red+Blue, 0,  Blue]
[0,    Red, 0,      Blue, 0]
[Red+Green, 0, All Four, 0, Blue+Yellow]
[0,    Green, 0,    Yellow,0]
[Green, 0, Green+Yellow, 0, Yellow]
```





Up-Convolution

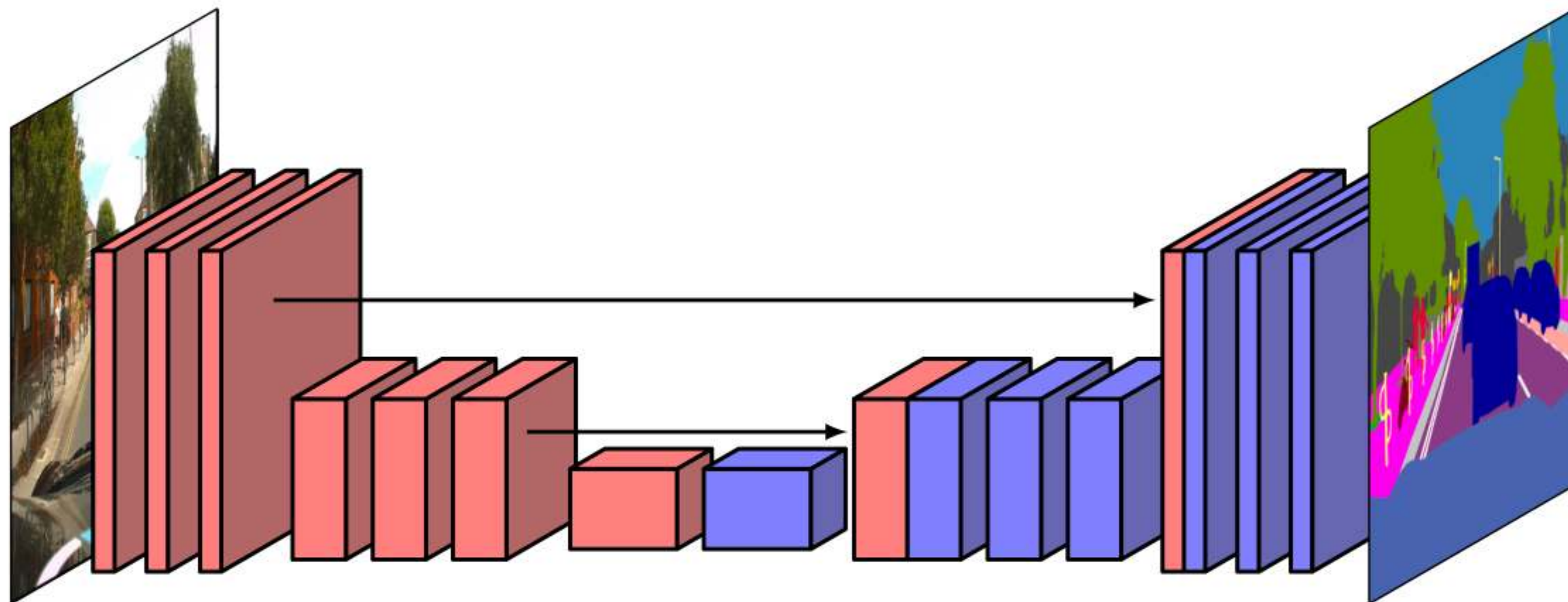




U-Net

access to higher-resolution spatial information via concatenating
down-sampling to up-sampling layers

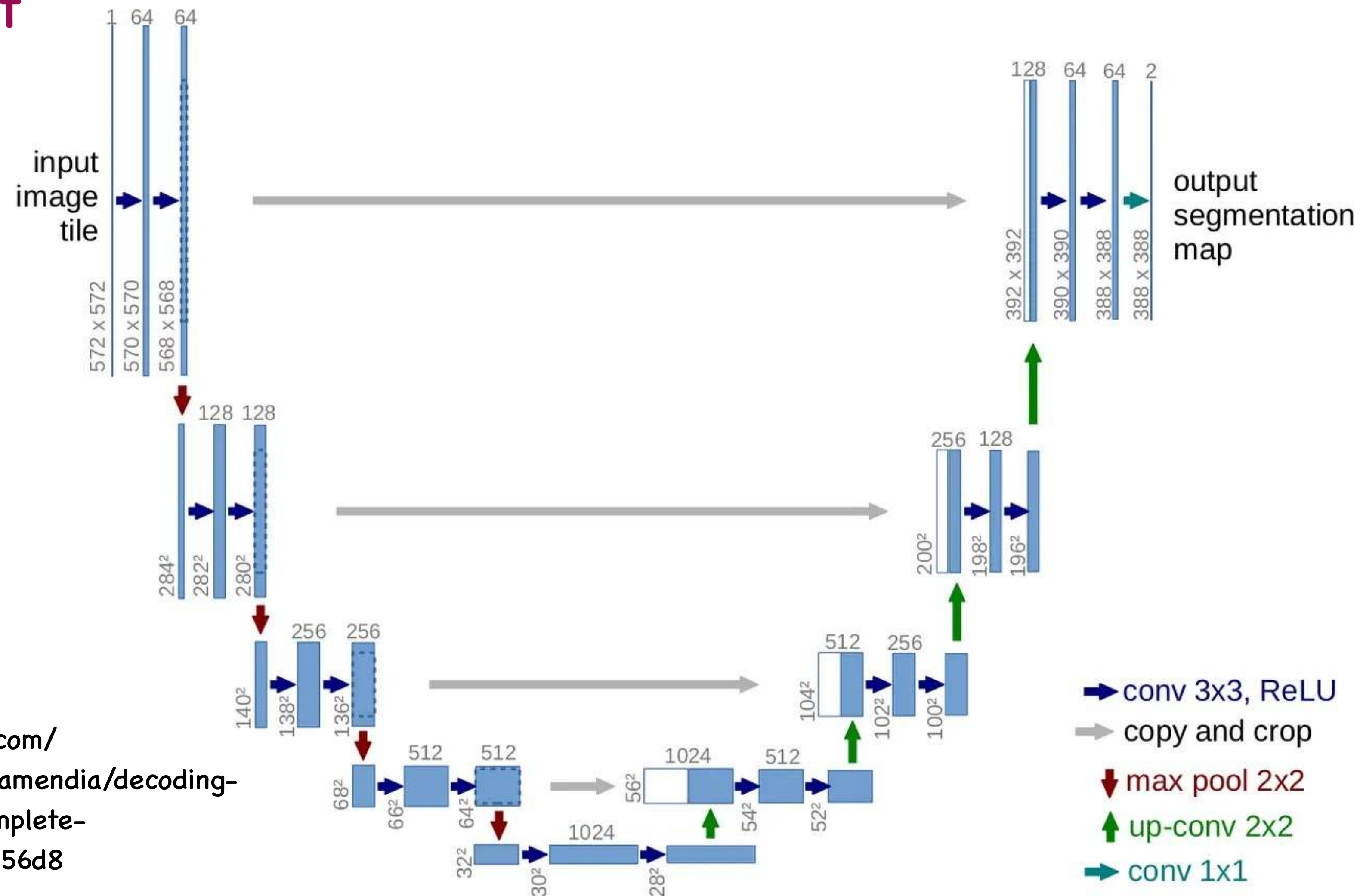
For each down-sampling layer there is a corresponding up-sampling layer





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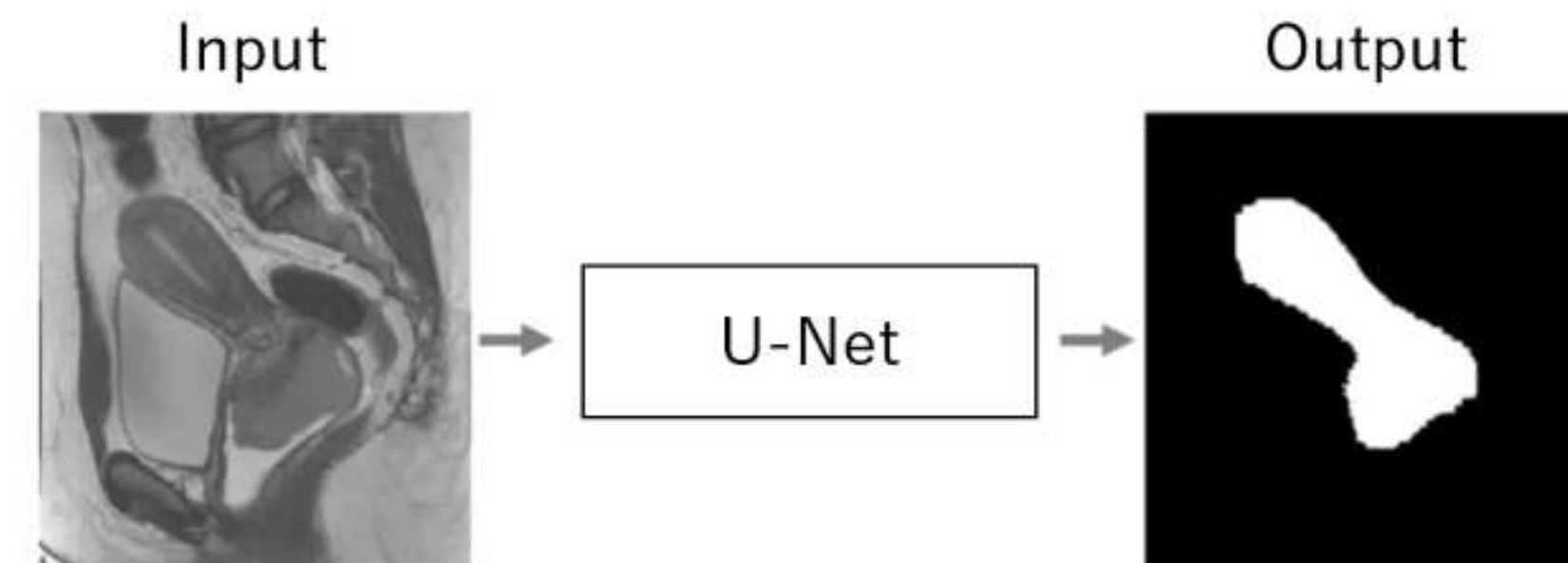
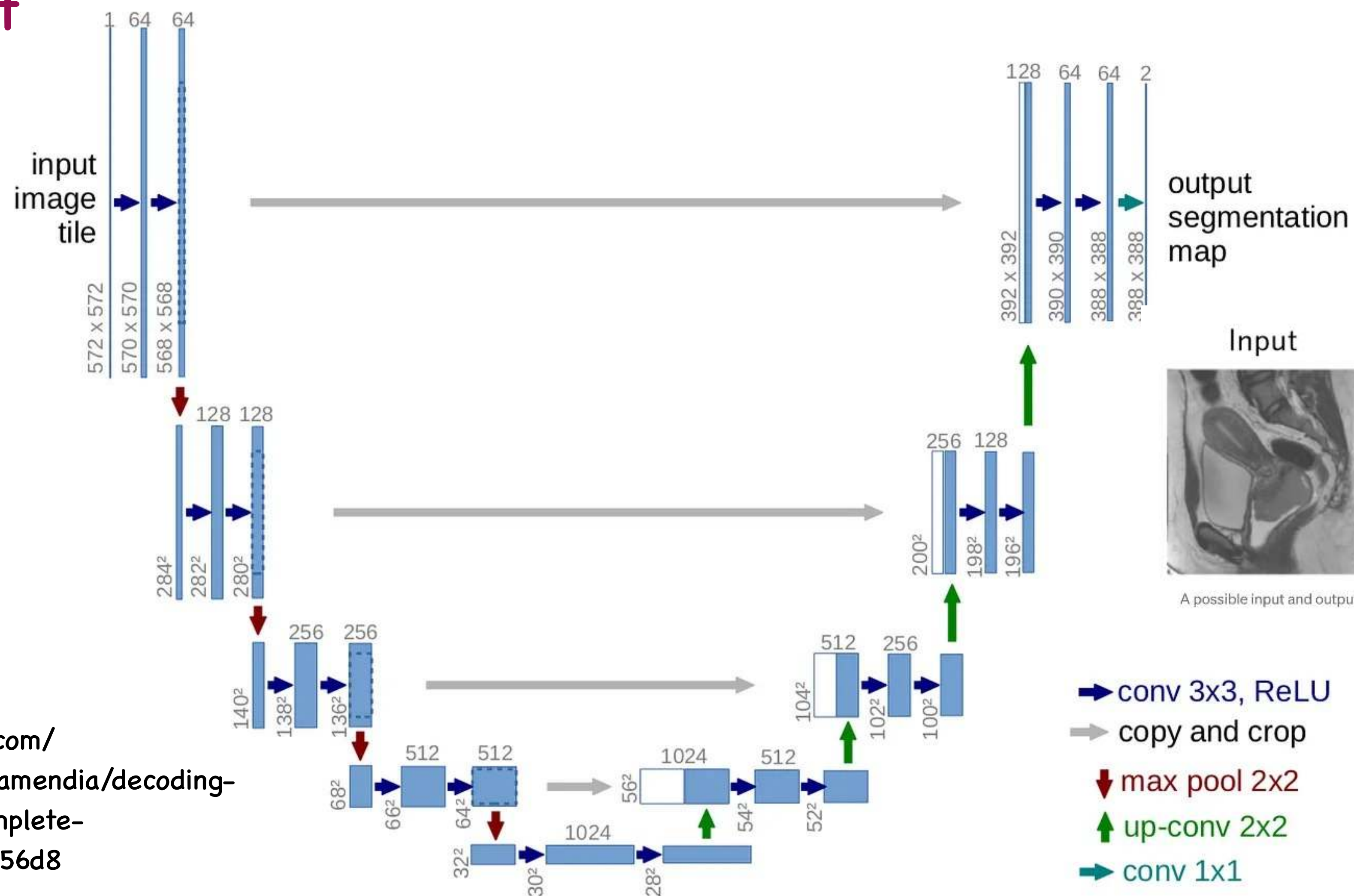
U-Net





AI-Driven HEP 15

U-Net



A possible input and output of a U-Net. In this case, a grayscale input and a single channel binary output.

<https://medium.com/@alejandro.itoaramendia/decoding-the-u-net-a-complete-guide-810b1c6d56d8>



AI-Driven HEP 15

Generalization Performance

increasing the number of layers in a network leads
to increase in generalization power significantly



Generalization Performance

increasing the number of layers in a network leads to increase in generalization power significantly

On the other hand increasing the layers leads to difficulties in training



Normalization

Vanishing gradients and Exploding gradients

$$\frac{\partial E}{\partial w_i} = \sum_m \cdots \sum_l \sum_j \frac{\partial z_m^{(1)}}{\partial w_i} \cdots \frac{\partial z_j^{(K)}}{\partial z_l^{(K-1)}} \frac{\partial E}{\partial z_j^{(K)}}$$

In a deep network multiplication of several positive or negative values makes troubles



Batch and Layer Normalization

$$\mu_n = \frac{1}{M} \sum_{i=1}^M a_{ni}$$

$$\sigma_n^2 = \frac{1}{M} \sum_{i=1}^M (a_{ni} - \mu_n)^2$$

$$\hat{a}_{ni} = \frac{a_{ni} - \mu_n}{\sqrt{\sigma_n^2 + \delta}}$$

$$\mu_i = \frac{1}{K} \sum_{n=1}^K a_{ni}$$

$$\sigma_i^2 = \frac{1}{K} \sum_{n=1}^K (a_{ni} - \mu_i)^2$$

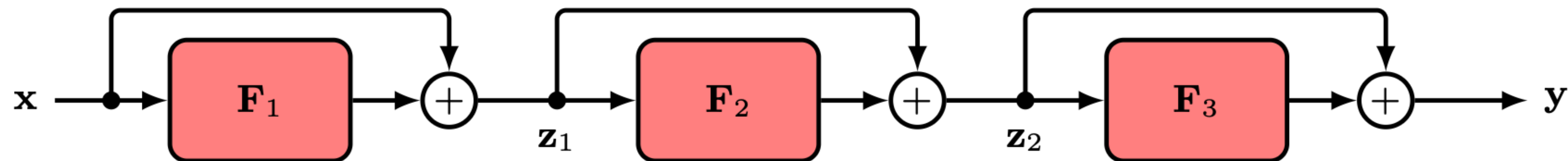
$$\hat{a}_{ni} = \frac{a_{ni} - \mu_i}{\sqrt{\sigma_i^2 + \delta}}$$

$$\tilde{a}_{ni} = \gamma_i \hat{a}_{ni} + \beta_i$$



Res-Net

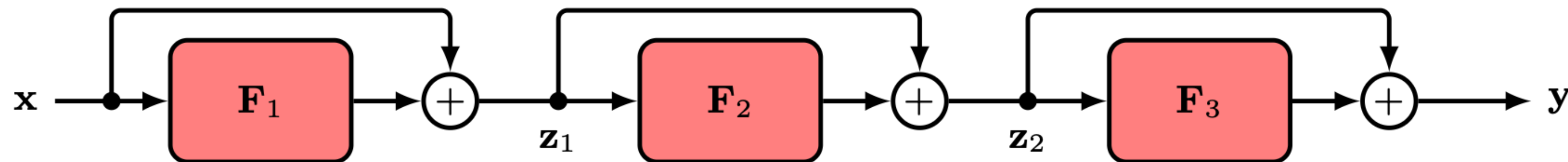
One solution for training very deep networks is using residual connections





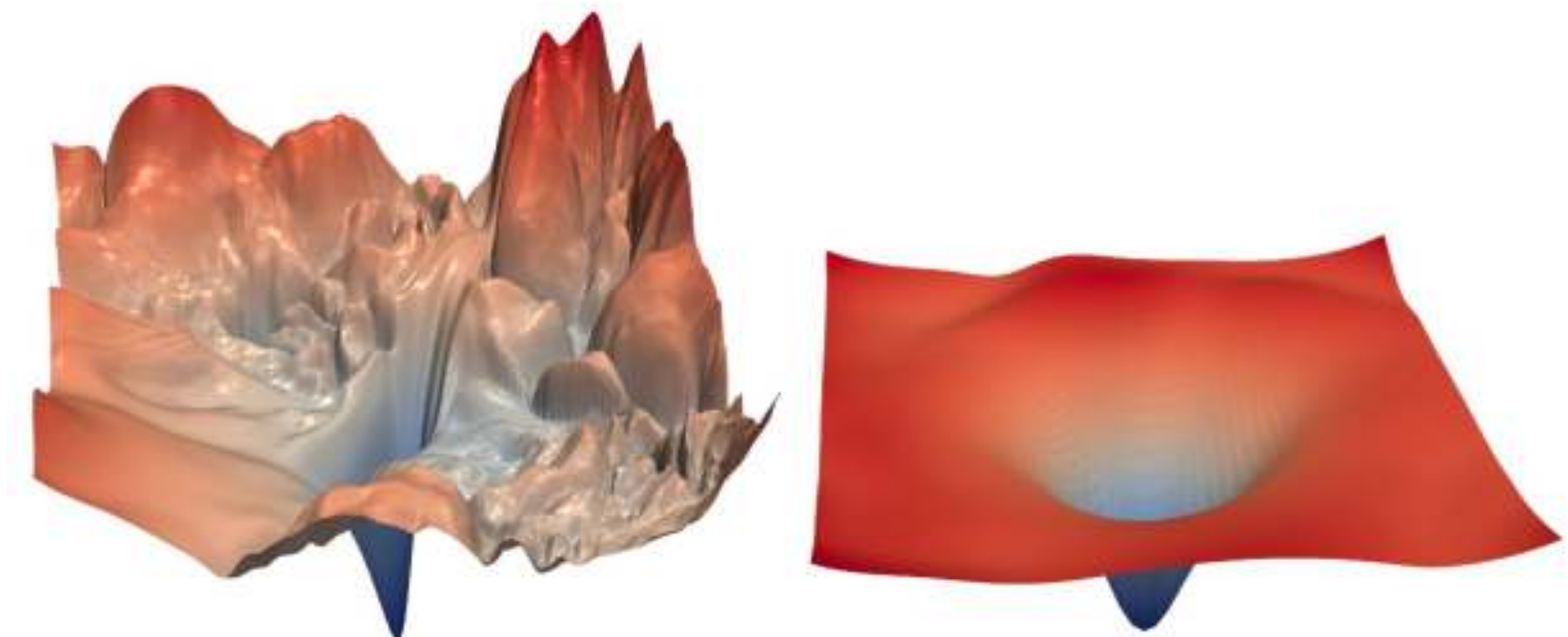
Res-Net

One solution for training very deep networks is using residual connections



In each block the function learns the residual between the identity map and the desired output

Create smoother error function surfaces





Generalization Error

how well a model generalizes

- 1) number of parameters
- 2) amount of training data
- 3) regularization and normalization
- 4) learning process



Generalization Error

bias and variance trade-off to minimize the generalization error

too few parameters $\rightarrow$ limited representational capacity (high bias) $\rightarrow$ high test error

as the number of parameters increases $\rightarrow$ the test error fall.

as the number of parameters is increased further $\rightarrow$ the test error increases again (high variance)



Learning Curves

Training data set

Validation data set

Test data set



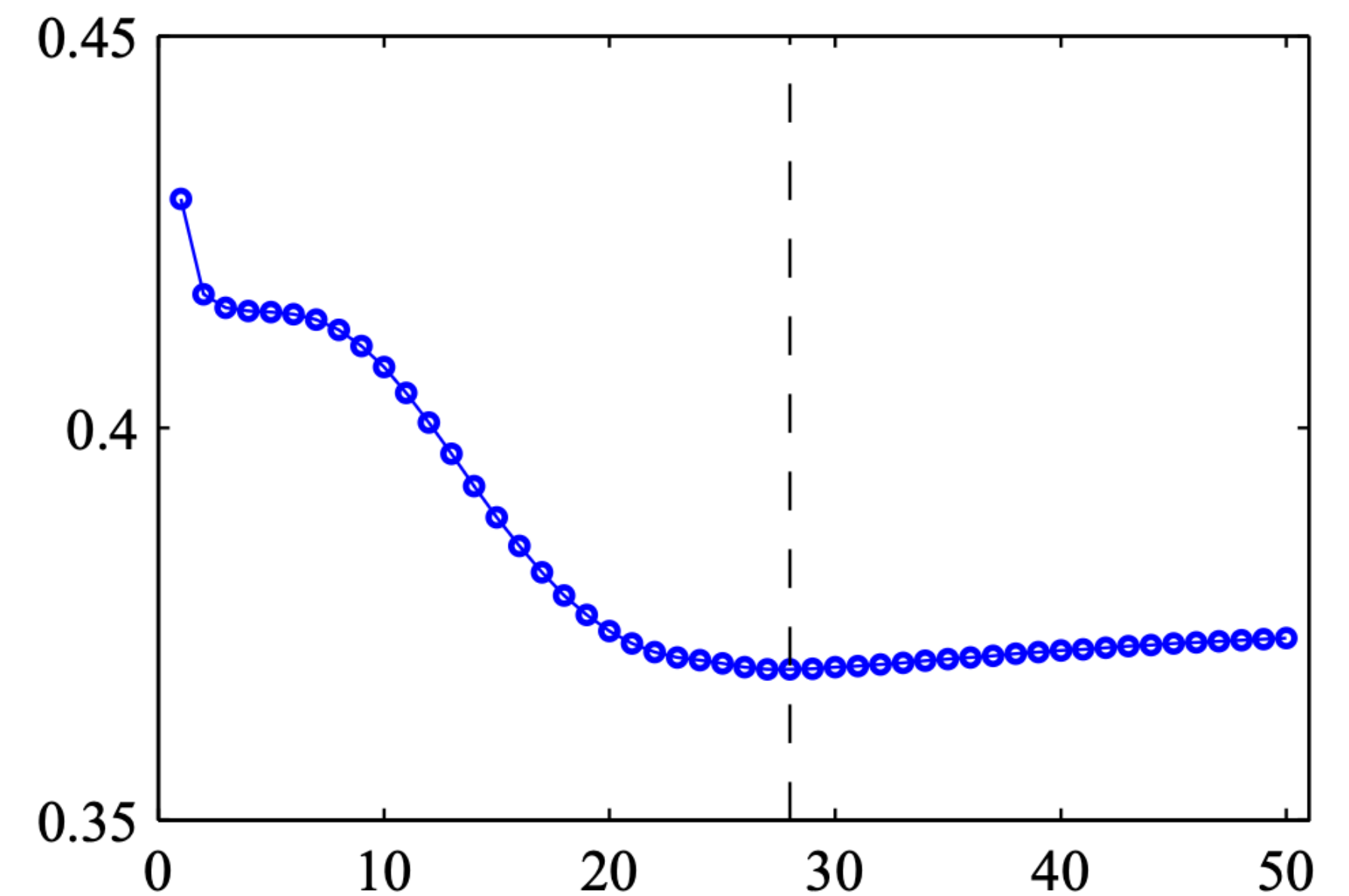
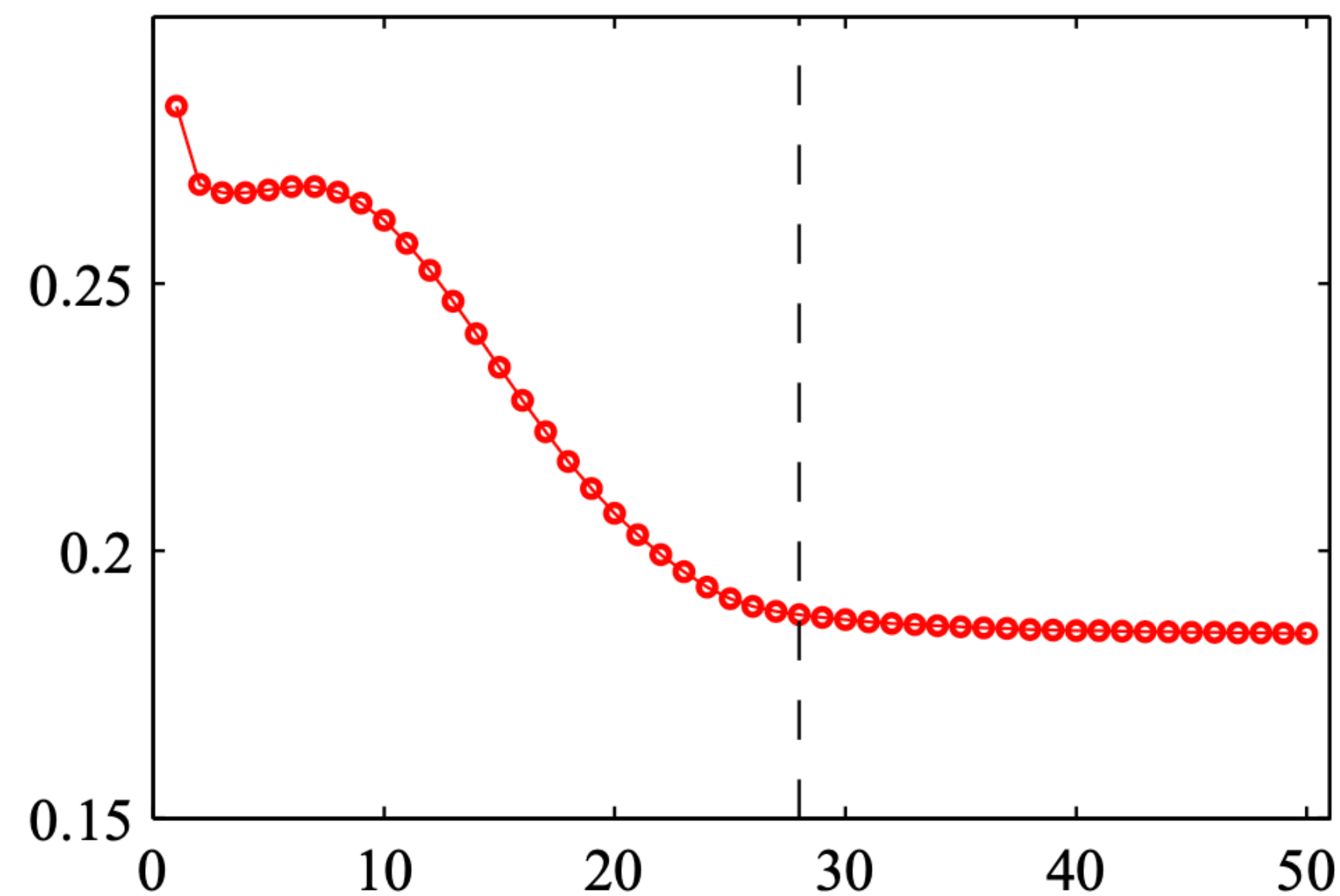
Learning Curves

Training data set

Validation data set

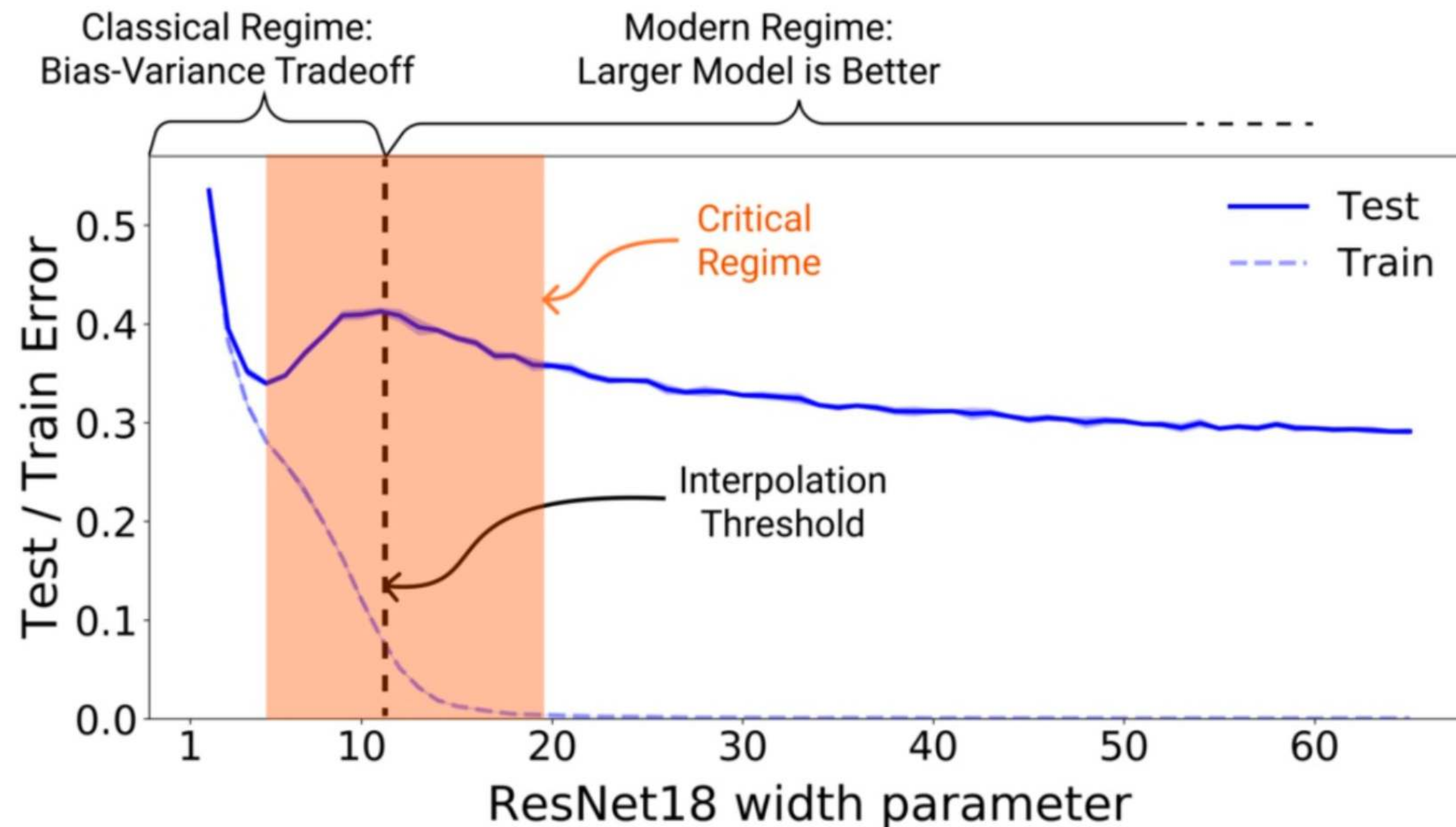
Test data set

good generalization performance





Double Descent

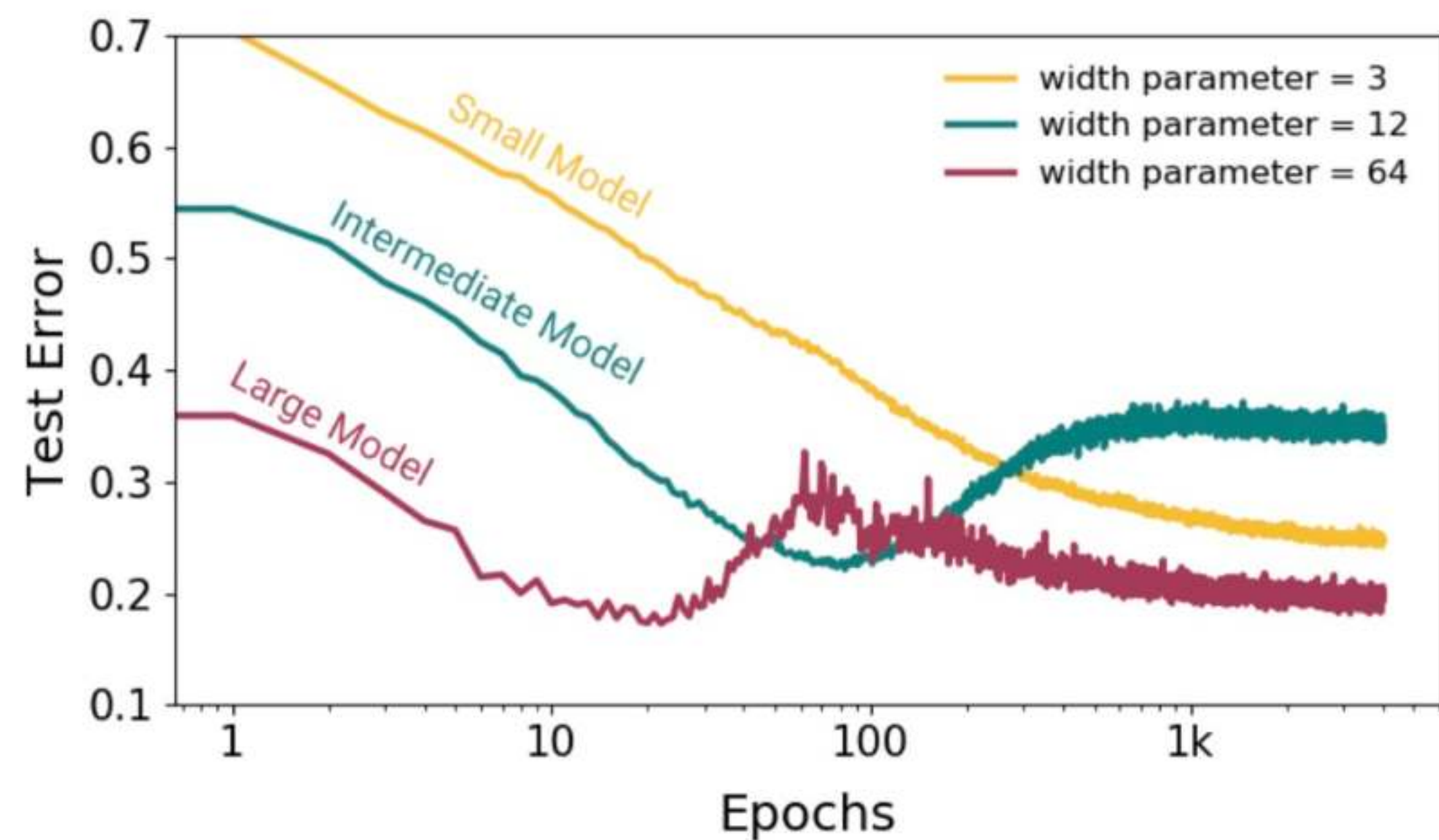


Width Parameters : The number of hidden units in each layer



Double Descent

Epochs Limitation



Sample Limitation

